Introduction to Cell Biology

Chapter 1: The Cell

What is a Cell?

A cell is the basic structural and functional unit of all living organisms. Cells are the smallest units of life that can replicate independently, and are often called the "building blocks of life."

Types of Cells

There are two main types of cells:

Prokaryotic Cells
- Lack a membrane-bound nucleus
- DNA floats freely in the cytoplasm
- Examples: Bacteria and Archaea
- Typically smaller (1-10 micrometers)

Eukaryotic Cells
- Have a membrane-bound nucleus
- DNA is organized into chromosomes
- Examples: Animal cells, plant cells, fungi
- Typically larger (10-100 micrometers)

Key Organelles

Mitochondria
The mitochondria are known as the "powerhouses" of the cell. They generate ATP (adenosine triphosphate) through cellular respiration, which provides energy for cellular processes.

Key features:
- Double membrane structure
- Contains its own DNA
- Site of aerobic respiration
- Produces approximately 36-38 ATP molecules per glucose molecule

Nucleus
The nucleus is the control center of the cell, containing the genetic material (DNA). It regulates gene expression and mediates the replication of DNA during the cell cycle.

Functions:
- Stores genetic information
- Controls cell activities
- Site of DNA replication
- Site of transcription (RNA synthesis)

Ribosomes

Ribosomes are the protein factories of the cell. They translate mRNA into proteins through a process called translation.

Characteristics:

- Can be free-floating or attached to endoplasmic reticulum
- Made of RNA and proteins
- Found in both prokaryotic and eukaryotic cells
- Do not have a membrane

Endoplasmic Reticulum (ER)

The ER is a network of membranous tubules and sacs involved in protein and lipid synthesis.

Two types:

- Rough ER: Has ribosomes attached; synthesizes proteins
- Smooth ER: No ribosomes; synthesizes lipids and detoxifies

Golgi Apparatus

The Golgi apparatus modifies, sorts, and packages proteins and lipids for delivery to targeted destinations.

Process:

1. Receives proteins from the ER
2. Modifies them (adds sugar groups)
3. Packages them into vesicles
4. Ships them to their final destination

Cell Membrane

The cell membrane (plasma membrane) is a selectively permeable barrier that separates the cell's interior from its external environment.

Structure:

- Phospholipid bilayer
- Embedded proteins
- Cholesterol molecules (in animal cells)
- Carbohydrate chains (glycoproteins and glycolipids)

Functions:

- Controls what enters and exits the cell
- Cell communication
- Cell recognition
- Maintains cell shape

Cell Cycle

The cell cycle is the series of events that take place in a cell leading to its division and duplication.

Phases:
1. G1 Phase (Gap 1): Cell growth and normal functions
2. S Phase (Synthesis): DNA replication occurs
3. G2 Phase (Gap 2): Preparation for cell division
4. M Phase (Mitosis): Cell division occurs

DNA Replication

DNA replication occurs during the S phase of the cell cycle. The process is semi-conservative, meaning each new DNA molecule contains one original strand and one newly synthesized strand.

Steps:
1. DNA helicase unwinds the double helix
2. DNA polymerase adds complementary nucleotides
3. Ligase seals the sugar-phosphate backbone
4. Two identical DNA molecules are produced

Photosynthesis vs. Cellular Respiration

These are complementary processes in the biosphere:

Photosynthesis (in plant cells):
- Converts light energy into chemical energy
- Occurs in chloroplasts
- Equation: $6CO_2 + 6H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6O_2$
- Produces glucose and oxygen

Cellular Respiration (in all cells):
- Converts chemical energy into ATP
- Occurs in mitochondria
- Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{ATP}$
- Produces carbon dioxide and water

Cell Differentiation

Cell differentiation is the process by which a less specialized cell becomes a more specialized cell type. This occurs during development and allows for the formation of different tissues and organs.

Examples:
- Stem cells → Blood cells
- Stem cells → Nerve cells
- Stem cells → Muscle cells

Key Vocabulary

- Cytoplasm: Gel-like substance inside the cell membrane
- Organelle: Specialized structure within a cell
- Homeostasis: Maintenance of stable internal conditions
- Osmosis: Movement of water across a membrane
- Diffusion: Movement of molecules from high to low concentration
- Active Transport: Movement of molecules against concentration gradient (requires energy)
- Passive Transport: Movement of molecules with concentration gradient (no energy required)

Important Concepts to Remember

1. All living things are made of cells
2. Cells come from pre-existing cells
3. The cell is the basic unit of life
4. Different cells have different functions
5. Structure and function are related in cells
6. Energy is required for all cellular processes
7. DNA contains genetic information
8. Proteins carry out most cell functions

Review Questions

1. What is the main difference between prokaryotic and eukaryotic cells?
2. What is the function of mitochondria?
3. Describe the process of DNA replication.
4. What happens during the S phase of the cell cycle?
5. How does the structure of the cell membrane relate to its function?

End of Chapter 1